

Kush Patel

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Education

Carnegie Mellon University

May 2027

Master of Science, Artificial Intelligence Engineering (GPA: 3.73/4.0)

Pittsburgh, PA

- **Coursework:** AI Systems & Toolchains, Machine Learning, Advanced NLP (PhD), Deep Learning, ML with Large Datasets, Trustworthy AI, Distributed Systems

Rutgers University - New Brunswick

Sep 2022 - Aug 2024

Bachelor of Science, Computer Science, Business Administration (Minor) (GPA: 3.67/4.0)

- **Coursework:** Computer Algorithms, Artificial Intelligence, Deep Learning, Operating Systems, Systems Programming

Technical Skills

Languages: Python, Go, C, TypeScript, SQL

AI/ML: LLMs, RAG, AI Agents, MCP, Transformers, QLoRA, Quantization, OCR, Computer Vision

Tools: FastAPI, Flask, Docker, Kubernetes, Git, CI/CD, AWS, GCP, Azure

Data & Systems: Apache Spark, Kafka, PostgreSQL, BigQuery, dbt, FAISS, Polars

Frameworks: PyTorch, TensorFlow, Hugging Face, JAX, LangChain, LangGraph, OpenCV, ONNX, YOLO

Experience

Advantest America, Inc | Software Engineer

Jun 2024 - Present

- Led development of a multi-agent AI system reading team emails and attachments through an LLM parsing pipeline and updating version-controlled project files, achieving 94% extraction accuracy
- Proposed and developed an OCR inspection pipeline with production team, tracking PCB layer stack-up and self-calibrating to each job size, saving an estimated \$1M+ annually in scrap and rework
- Designed ETL pipelines using Apache Spark and dbt with CI/CD-automated testing to clean and transform 730M+ records, powering production reporting and analytics at 99% uptime
- Automated 80% of the inspection workflow by connecting robotic arms to legacy measurement software via a TCP/IP communication bridge
- Engineered and deployed a scalable FastAPI server with caching and optimized data processing, powering a real-time dashboard across 5 regions for live manufacturing data

Technology Intern

Aug 2022 - May 2024

- Onboarded and supported 1,000+ users through Azure AD setup, MFA rollout, and Intune deployment, writing guides that cut repeat support requests
- Automated audits and backup verification across 1,000+ endpoints, streamlining compliance reporting

Projects and Research

Efficient LLM Reasoning with Early Exit | NLP Research [Report](#)

2026

- Trained an MLP probe on 4,096-dim Llama-3.1-8B hidden states to predict answer readiness mid-reasoning (0.82 ROC-AUC), a lightweight alternative to reward models
- Reduced simulated inference tokens by up to 98.1% with a step-level early-exit policy, outperforming token-probability baselines on four benchmarks

Prompt Injection Defense for LLM Agents | LLM Security, AI Agents [Report](#)

2026

- Built a 3-layer guardrail pipeline for LLM agents using structural delimiters, a router LLM firewall, and an LLM-as-a-judge, cutting prompt injection success on InjecAgent from 75.0% to 2.3%
- Added a deterministic symbolic verifier and TF-IDF RAG threat intelligence to catch attacks the LLM judge missed, intercepting 97.7% of residual failures

RAG Question-Answering System | Retrieval, Re-ranking, Web Crawling [GitHub](#)

2026

- Developed a data pipeline to crawl, clean, and chunk web pages on CMU and Pittsburgh culture into a searchable knowledge base for question answering
- Improved F1 by 10% over BM25 by fusing dense MiniLM/FAISS retrieval with BM25 via RRF and adding cross-encoder re-ranking to ground Mistral-7B

Retail Shelf Gap & Product Count Detection | DL Research, Computer Vision, Data Modeling [Report](#)

2026

- Generated a 5,000+ image synthetic gap dataset from UniK3D metric depth and hand-labeled a real-world benchmark 10K+ products, 1.3K+ gaps
- Collaborated in team of 4 and innovated RGB-D fusion models FCOS, Depth Anything, Faster RCNN, Unik3D and fine-tuned YOLOv11 and RT-DETR, predicting product counts at 14.5 RMSE